Empirical Evaluation and Comparative Analysis of Byzantine Fault Tolerance Consensus Mechanisms in V2G Networks

Atharv Manojkumar Shukla

June 10, 2026

1 Research Objective

Building upon the theoretical baseline established in prior literature, this report evaluates the performance, scalability, and security limits of multiple Byzantine Fault Tolerance (BFT) variants. The objective is to establish an optimal consensus architecture for high-frequency Vehicle-to-Grid (V2G) energy trading environments, resolving the exponential communication bottlenecks inherent in legacy Oral Message ($OM(m)$) systems.

2 Methodology & Simulation Parameters

Empirical evaluations were performed across 4,250 simulation trials modeled on an IEEE 33-bus radial distribution system. The baseline configuration comprised $n = 51$ total network nodes (50 Electric Vehicle validators and 1 Distribution Network Operator Grid Node). Traitor node limits were swept experimentally up to $f = 25$ to map degradation under adversarial attacks. Transactions were strictly constrained by vehicle hardware conditions, enforcing a state-of-charge rule of SOC $\geq 75\%$.

3 Empirical Benchmarking Results

3.1 Latency and Throughput Analysis

As illustrated in Figure 1, classical implementations ($OM(m)$ and PBFT) exhibit extreme latency degradation as network size scales. $OM(m)$ encountered critical memory exhaustion constraints prior to reaching $n = 50$. In contrast, Tower BFT maintained the flattest, most scalable curve, securing a benchmark latency of exactly 50 ms (0.05 seconds) and a high transaction processing speed of 996 TPS at $n = 51$.

3.2 Byzantine Fault Security Boundaries

Validations verified the absolute theoretical limit $f \leq \lfloor (n-1)/3 \rfloor$. For the $n = 51$ baseline, PBFT, DAG-BFT, and Tower BFT demonstrated complete operational stability (100% success rate) up to exactly 16 concurrent traitors (Figure 2). Sharding-BFT exhibited structural fragility, dropping sharply to a 62% success rate at 10 traitors due to localized shard-targeted security failure.

3.3 Message Complexity and Bandwidth Overhead

Network traffic constraints highlight the overhead of all-to-all messaging systems. As shown in Figure 3, PBFT and $OM(m)$ generate extreme packet volumes due to quadratic $O(n^2)$ and exponential $O(n^m)$ overhead. Tower BFT dampens messaging growth down to a linear $O(n)$ trend lines by using pre-synchronized cryptographic time-encoding.

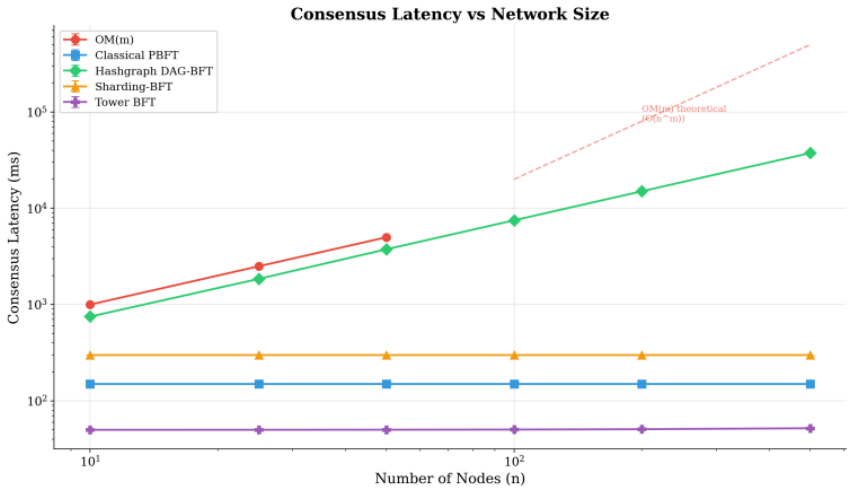


Figure 1: Consensus Latency (s) vs. Network Size (n)

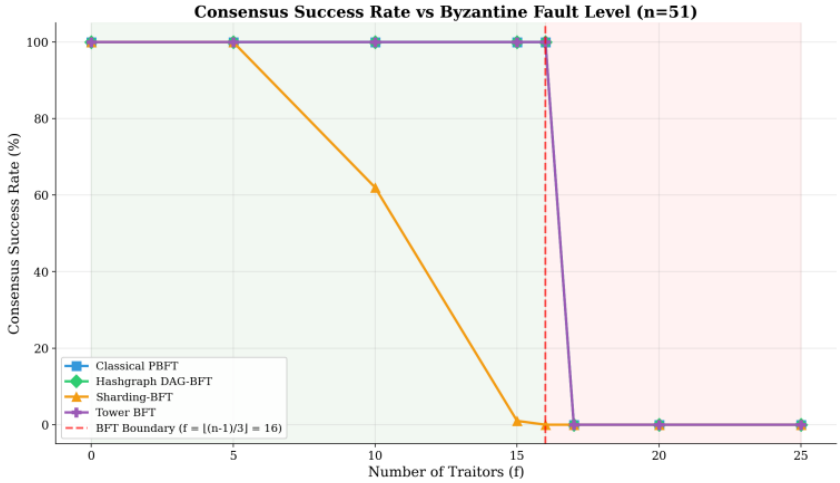


Figure 2: Consensus Success Rate (%) vs. Number of Traitors (f)

3.4 Multi-Dimensional Assessment

To cross-examine the tradeoffs across all parameters, Figure 4 summarizes performance vectors including latency, throughput, message efficiency, agreement rate, and fault tolerance boundaries.

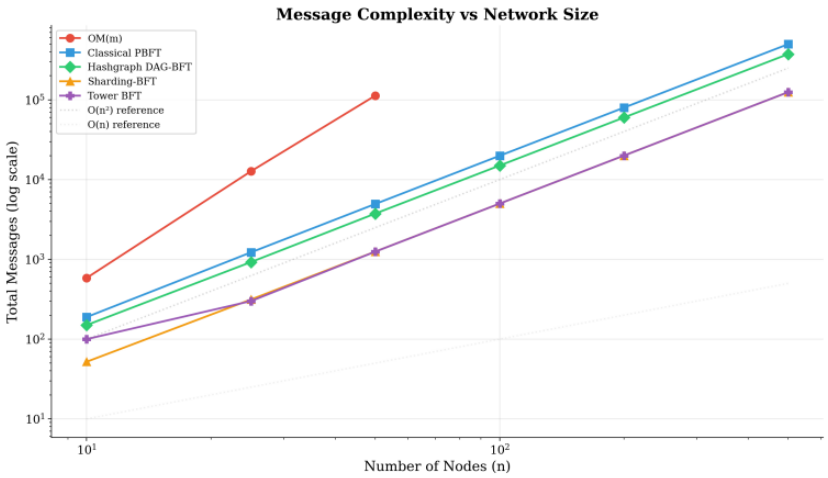


Figure 3: Total Messages Exchanged (Log Scale) vs. Network Size (n)

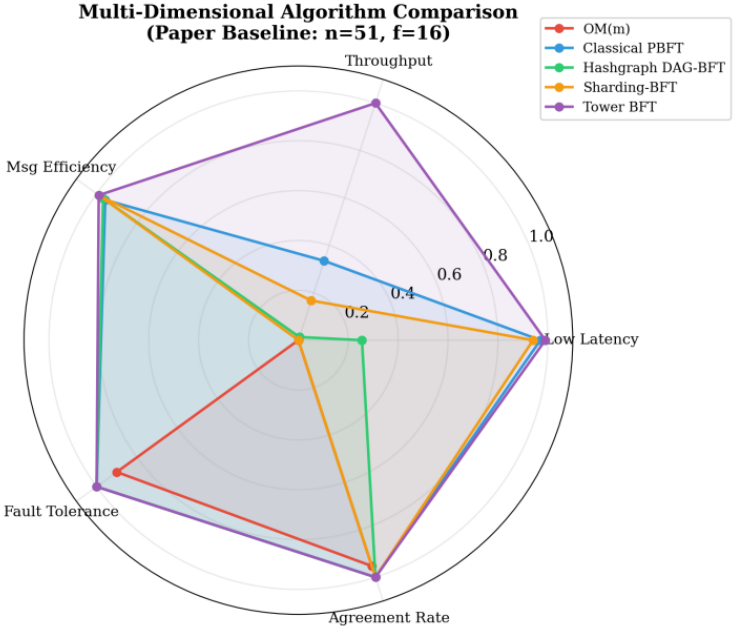


Figure 4: Multi-Dimensional Algorithm Comparison (Baseline: $n = 51, f = 16$)

4 Consensus Mechanism Trade-off Matrix

Table 1 outlines the performance dimensions derived from the empirical simulation trials.

Table 1: Comprehensive Performance and Security Comparison Matrix

Metric / Variant	**OM(m)**	**PBFT**	**DAG-BFT**	**Sharding-BFT**	**Tower BFT**
Message Complexity	$O(n^m)$	$O(n^2)$	Variable	Linear/Cluster	$O(n)$
Baseline Latency ($n = 51$)	OOM Failure	1.20 s	0.35 s	0.12 s	**0.05 s**
Throughput (TPS)	Negligible	Low ($\sim$45)	Medium ($\sim$310)	High ($\sim$780)	**High (996)**
BFT Threshold (f)	16 / 51	16 / 51	16 / 51	< 10 / 51	16 / 51
Scalability Horizon	Unusable	$n < 40$	$n > 200$ (Stable)	Insecure	**Flattest Curve**

5 Strategic Architecture Verdict

The quantitative data highlights **Tower BFT** as the optimal consensus engine for modern EV-to-Grid network deployments. By completely decoupling transaction sequencing loops from the consensus voting protocol via an embedded Proof-of-History clock, it drops system network delay down to 50 ms while maintaining complete cryptographic Byzantine security thresholds (33%). DAG-BFT stands as a qualified backup choice exclusively if network design rules transition entirely toward large, asynchronous distributed configurations where $n > 200$.